



American International University-Bangladesh (AIUB)

**Faculty of Science and Technology (FST)
Department of Computer Science (CS)**

SDPM Group Project, Spring 2022-23

Project Title: One-stop Home Management Solutions

Section: 'C'

Submitted by:

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1.0 Introduction

We are building an application that can solve all basic problems that may arise in a house. This system will help all those busy people who can't give enough time to the maintenance of their house because of a busy schedule. From electricians to house-shifting laborers, from plumbers to painters, users can hire several types of helping hands as needed. We will provide experienced electricians, plumbers, and laborers at our users' doorstep. We will also offer a feature to hire a van or truck if the user is shifting furniture from one place to another. Users will select the number of laborers needed and create a list of tasks to do to proceed. Using this system will be very useful to manage all these works while saving time, cost, and labor.

2.0 Project Title

One-stop Home Management Solutions.

3.0 Objectives

The objectives of developing the system are the following-

- Develop a comprehensive and user-friendly application that enables busy individuals to quickly and easily book and manage a range of home maintenance services.
- Provide a range of experienced professionals for different tasks such as electricians, plumbers, painters, and laborers.
- Offer the ability to book van or truck services for furniture shifting.
- Allow users to select the number of laborers required for the job and create a list of tasks to be completed.
- Implement a robust system for managing appointments, scheduling tasks, and tracking progress.
- Continuously improve the platform based on user feedback and market research.

4.0 Justification

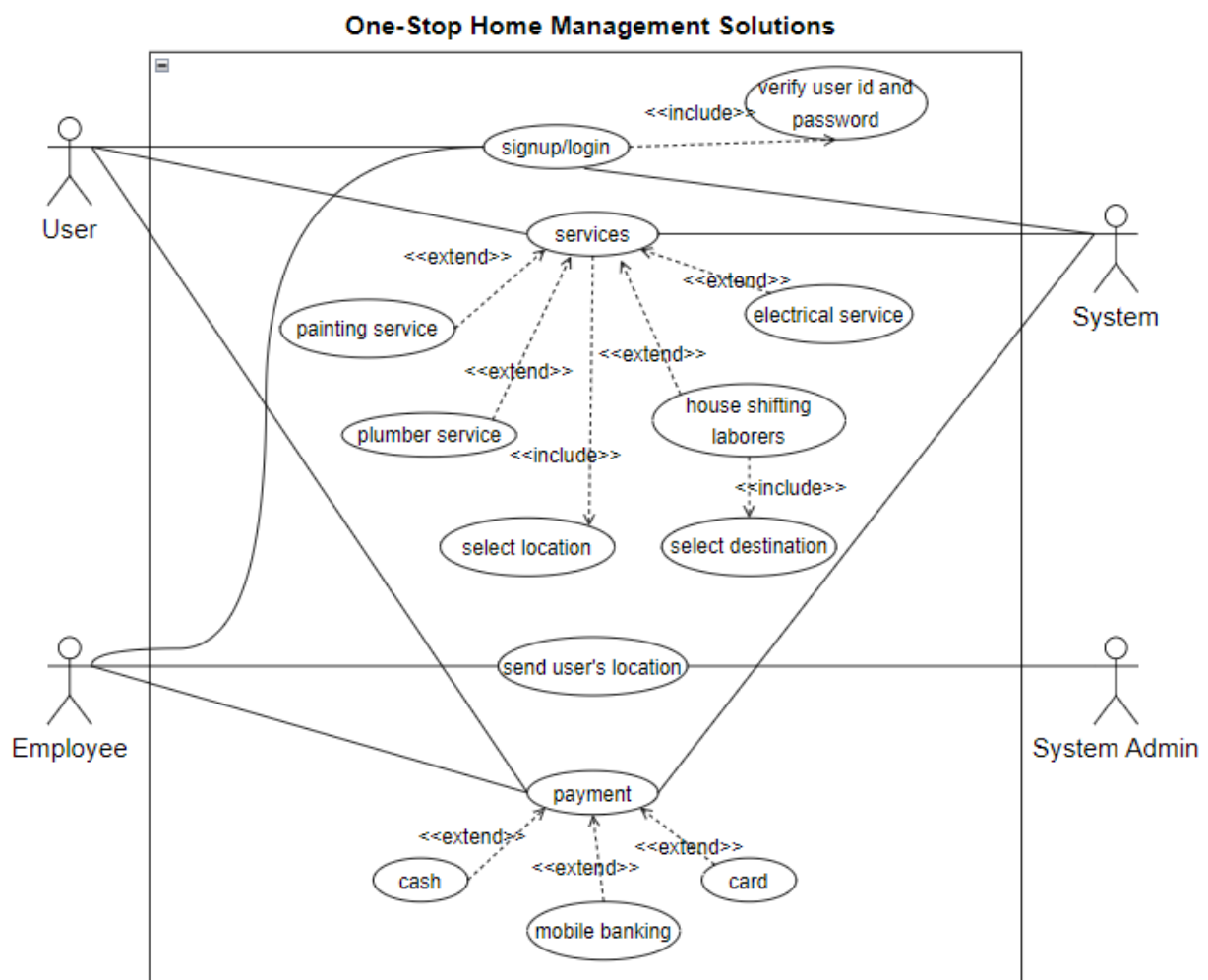
This system will benefit all busy people who don't have enough time to do housework due to a hectic schedule. From electricians to house movers, plumbers to painters, users can hire a variety of assistance as needed. For example, if a user needs to repair their washing machine, we will send a washing machine expert; if a user needs to set up their oven, we will send a skilled electrician. This app will be extremely useful for people who spend the majority of their time outside.

5.0 Systems Overview (Includes Use case diagram)

The system is designed to save users time, cost, and effort by providing a one-stop shop for all their household maintenance needs. The platform provides a transparent pricing system that is affordable and competitive, and all service providers are fully vetted and qualified professionals. The system also offers a robust appointment management system that allows users to easily schedule and manage their appointments.

Overall, the system aims to simplify and streamline the process of managing household chores and maintenance for busy individuals, providing a reliable and hassle-free solution to their daily needs.

Here is the use case diagram of the proposed system-



6.0 Stakeholders Analysis

For the development of this project, the following types of stakeholders have been identified-

Primary Stakeholders: These are stakeholders who have a direct and significant interest in the project's outcome and can directly influence its success.

1. End-Users
2. House Maintenance Service Providers
3. Van/Truck Rental Service Providers
4. Project Team

Secondary Stakeholders: These stakeholders may be indirectly affected by the project but still have an interest or involvement.

1. Executives and Senior Leaders
2. Project Sponsors/Investors
3. Payment Gateway Providers (SSLCommerz, Payoneer, etc.)

Internal Stakeholders: These are individuals or groups within the organization directly affected by the project.

1. Project Managers
2. Project Team
3. Testing Team
4. Executives and Directors

External Stakeholders: These are individuals or groups outside the organization who have an interest in or are impacted by the project

1. End-Users
2. Payment Gateway Providers
3. Project Sponsors/Investors

Negative Stakeholders: Negative stakeholders are individuals, groups, or entities who are adversely affected by or have concerns, objections, or opposing interests regarding a project or initiative.

1. Competitors
2. Existing Service Providers
3. Local Contractors

7.0 Feasibility Study

Technical Feasibility: The technical feasibility of the project appears to be promising due to the following factors:

- **Availability of resources:** As long as resources such as skilled electricians, plumbers, laborers, and rental services are available and accessible, the system can effectively connect users with the required services.
- **Technology infrastructure:** Modern development tools and frameworks enable the creation of user-friendly interfaces and seamless integration.
- **Scalability:** The system can handle a growing user base and increasing service demands using scalable architecture and cloud solutions.
- **Security:** Robust security measures, encryption, user authentication, and secure payment gateways can protect user data and financial transactions.
- **Integration:** The system can integrate with service providers, payment gateways, and external systems through proper APIs and data exchange protocols.

Financial Feasibility:

- **Revenue Generation:** The system offers a range of services and by charging a fee for each service or implementing a commission-based model, the system can generate revenue from transactions facilitated through the platform.
- **Market Demand:** The system targets busy individuals who struggle with house maintenance due to time constraints. There is a potential market demand for such services, especially in urban areas with a high population of working professionals.
- **Return on Investment (ROI):** Assessing the projected return on investment is crucial for financial feasibility. This includes estimating the development and operational costs, projected revenues, and potential profitability over a defined period.

8.0 Systems Component

UI Components:

1. Login Interface
2. Registration Interface
3. Service Selection and Booking Interface
4. Task Creation and Management Interface
5. Interactive Map Interface
6. User Profile Interface
7. Payment Interface
8. Communication Interface
9. Feedback and Rating Interface

Business Logic Components:

1. User Authentication
2. Service Provider Matching and Scheduling
3. Task Assignment and Tracking
4. Payment Processing

Data Access Components:

1. All Database System

9.0 Process Model to be Followed

The chosen process model for this project is **Agile**, a dynamic and iterative approach to software development. Agile methodologies have surpassed traditional waterfall-model processes by becoming more flexible, rapid, lean, responsive, and consistent. In Agile, development is divided into short iterations which are called "sprints." Each sprint focuses on delivering a set of prioritized features. The project team collaboratively plans and executes these sprints.

The Agile process fosters transparency and encourages close involvement of stakeholders throughout the development cycle. It allows for continuous refinement and adjustment of the product based on user feedback, ensuring that the final solution meets the desired objectives.

An agile model can be implemented in this project by establishing principles, defining a prioritized product backlog, planning sprints, holding daily meetings, iterating feature implementation and testing, conducting reviews and retrospectives, adapting based on feedback, and fostering collaboration. This approach ensures efficient and customer-centric development, reducing risks and enhancing project success.

10.0 Efforts Estimation

To estimate the required effort using bottom-up estimation, we utilized the Work Breakdown Structure (WBS) technique, which involved breaking down the project's activities into smaller, manageable tasks. The resulting WBS outline includes the following list of tasks that need to be completed for successful project execution:

- Planning
- Requirements Gathering
- Design
- Development
- Testing
- Documentation

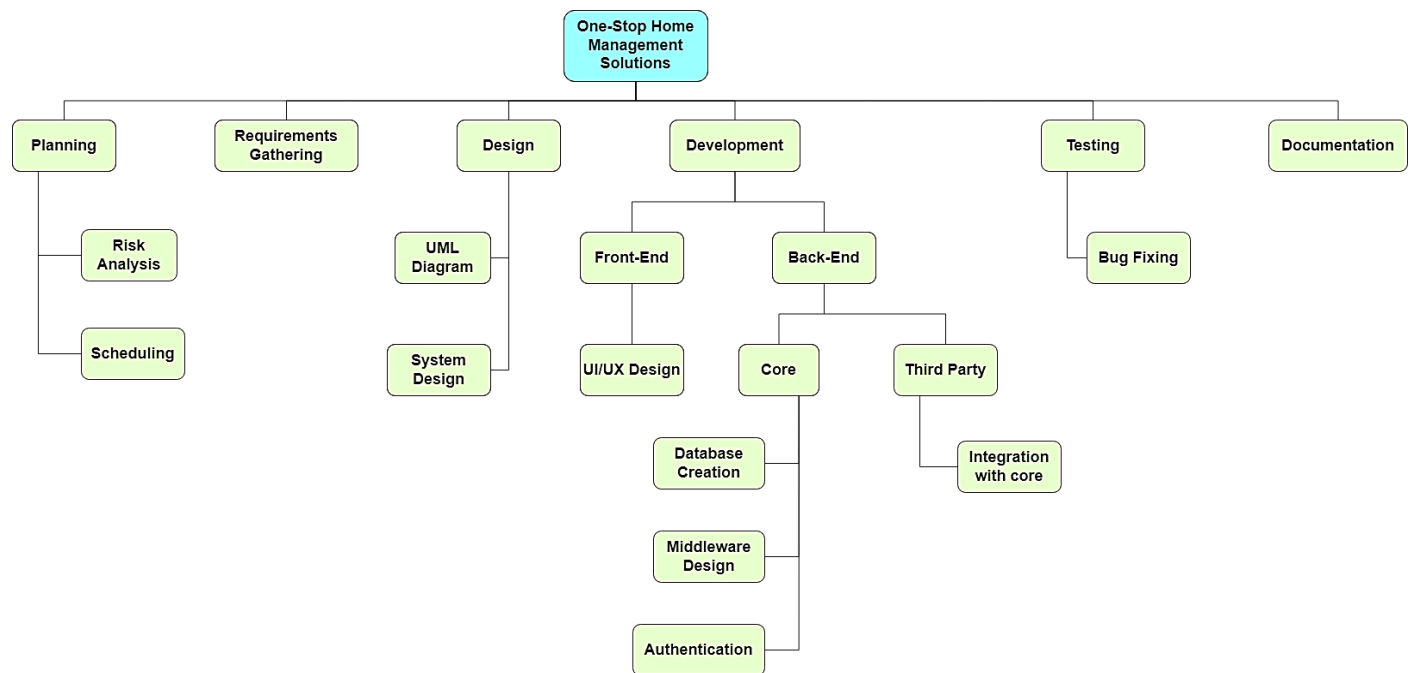


Fig: Work Breakdown Structure (WBS)

TASKS	SLOC
Functional Modules	11,000
UI/UX Design	5,000
API Integration	2,000
Database Creation	4,500
Authentication	2,500
Middleware Design	4,000
Integration with core modules	3,000
Total	32,000

According to Cost Constructive Model (COCOMO),
Estimated Source Line of Code (SLOC)= 32,000

PM: person-months needed for the project

SLOC: source lines of code

P: project complexity

DM: duration time in months for the project

T: SLOC-dependent coefficient

ST: average staffing necessary

Coefficient<EffortFactor> = 2.4

P = 1.05

T = 0.38

Effort = PM = Coefficient<EffortFactor>*(SLOC/1000)^P

= 2.4 * (32,000/1000)^{1.05}

= 91.33 person-months

Development Time = DM = 2.50*(PM)^T

= 2.50*(91.33)^{0.38}

= 13.90 months ≈ 14 months

Required Number of people = ST = Effort (PM)/Development Time (DM)

= 91.33/13.90

= 6.57 people ≈ 7 people

From the above estimation,

Project Type	Coefficient<EffortFactor>	P	T
Organic	2.4	1.05	0.38

11.0 Activity Diagram

Precedence Table:

Label	Activity	Precedence	Duration (Weeks)
A	Software Selection	-	2
B	Planning	-	2
C	Requirements Gathering	B	4
D	Design	C	5
E	Development & Testing	A, D	20
F	Deployment	E	2
G	User Testing	F	4
H	Debugging & Improvements	G	6
I	Documentation	G	8

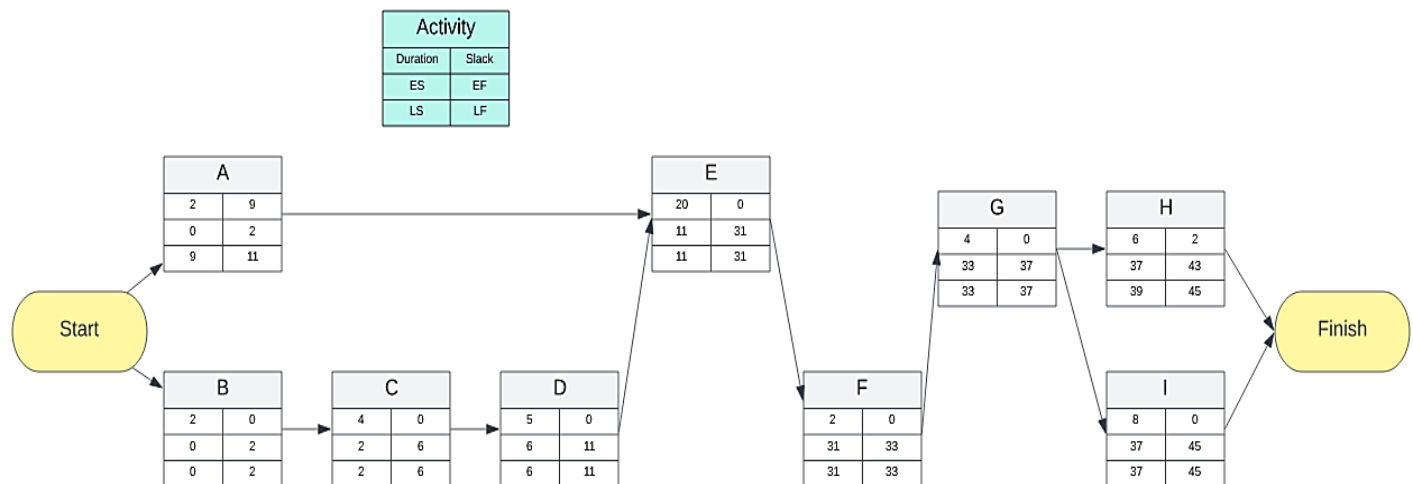


Fig: Activity Network Diagram

Critical Path Analysis:

Path-1: A – E – F – G – H = 34 weeks

Path-2: A – E – F – G – I = 36 weeks

Path-3: B – C – D – E – F – G – H = 43 weeks

Path-4: B – C – D – E – F – G – I = 45 weeks

Here, Path-4 is the critical path as it is the longest path in the diagram.

12.0 Risk Analysis

Some potential risks associated with this project include the following:

- R-1 Resource Constraints:** Insufficient, or limited availability of skilled personnel, equipment, or infrastructure may impact the project's progress and quality.
- R-2 Scope Creep:** The project requirements may expand or change during the development process, leading to scope creep. This can result in delays, increased costs, and potential conflicts with the project timeline.
- R-3 Staff Turnover:** When experienced team members leave, valuable knowledge, expertise, and project-specific information may be lost. This can lead to delays, decreased productivity, and difficulties in maintaining project continuity.
- R-4 Late Delivery of Deliverables:** Late delivery can result in missed deadlines, increased costs, customer dissatisfaction, and inefficient resource utilization, potentially impacting the project's success and reputation.
- R-5 Incorrect Estimation:** Incorrect estimation can lead to schedule delays, budget overruns, scope creep, and compromised quality.
- R-6 Fewer users in Beta Testing:** Having fewer users than estimated can result in limited feedback and reduced coverage of potential issues. This may lead to a higher risk of undiscovered bugs, usability issues, and insufficient validation of the software's performance in real-world scenarios.
- R-7 Vendor Reliability Issues:** Dependency on external vendors, such as laborers or transportation, etc. may introduce risks related to their reliability, availability, or quality of service.

Potential Risks	Likelihood	Impact	Risk Exposure (RE)
Resource Constraints	5	7	35
Scope Creep	4	6	24
Staff Turnover	3	7	21
Late Delivery of Deliverables	4	5	20
Incorrect Estimation	4	8	32
Fewer users in Beta Testing	6	5	30
Vendor Reliability Issues	5	6	30

Here, Likelihood and impact are assessed on a scale of 1 to 10, with higher ratings indicating more severe hazards and lower ratings representing less significant risks.

13.0 Budget for the Project

Development Cost:

Total working days = Monthly working days * DM = $22 * 14 = 308$ days

Total Working hours = Working hours per day * Total working days = $6 * 308 = 1848$ hours

Development cost per employee = Total Working hours * per hour salary [per hour salary = 200]
 $= 1848 * 200 = 3,69,600$ BDT

Total development cost = ST * Development cost per employee = $7 * 369600$
 $= 25,87,200$ BDT

Project manager salary for 14 months = $(14 * 20,000) = 2,80,000$ BDT

Management and other employees' salary for 14 months = $3,00,000$ BDT

UI/UX Designer Salary for 14 months = $(15,000 * 14 * 1) = 2,10,000$ BDT

Database Administrator Salary for 14 months = $(15,000 * 14 * 1) = 2,10,000$ BDT

Maintenance Cost:

Monthly 12 Hours (6 months and per hour salary 1000) Cost = $(12 * 6 * 1000) = 72,000$ BDT

Hardware Expenses:

Server cost = $1,20,000$ BDT

Desktop/Laptop cost = $(7 * 60,000) = 4,20,000$ BDT

Printer cost = $20,000$ BDT

Total hardware expenses = $(120,000 + 420,000 + 20,000) = 5,60,000$ BDT

Other Expenses:

Office rent for 14 months = $(14 * 15,000) = 2,10,000$ BDT

Utility bills = $2,00,000$ BDT

Stationaries cost per month = $2,000$ BDT

Total cost for 14 months = $(2,000 * 14) = 28,000$ BDT

Total Estimated cost = $(25,87,200 + 2,80,000 + 3,00,000 + 2,10,000 + 2,10,000 + 72,000 + 5,60,000 + 2,10,000 + 2,00,000 + 28,000) = 46,57,200$ BDT

25% profit of total estimated cost = $(0.25 * 46,57,200) = 11,64,300$ BDT

Total estimated budget = $(46,57,200 + 11,64,300) = 58,21,500$ BDT

14.0 Conclusion

In conclusion, our project aims to revolutionize the way people manage the maintenance of their homes by providing a comprehensive solution to address all fundamental household problems. With a user-friendly application, experienced professionals at users' doorsteps, and convenient features like hiring van/truck, we empower busy individuals to effortlessly handle their home maintenance needs. By embracing this innovative system, users can save valuable time, reduce costs, and eliminate the hassle of coordinating multiple services. This system is a well-timed application to launch in the market. This system is very much needed in a busy city like Dhaka. So, we already can expect a vast number of users to take benefits from our system.